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EBOOK

The Four Essentials of Applied Intelligence

Solving the data-AI disconnect in retail and CPG to protect margins and drive growth with Azure Databricks



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Retail's AI requirements have changed again

Unlike other industries, retailers don't have the luxury to wait for AI to settle in as a comfortable tool. Pressure to innovate has increased throughout the industry. Any customer can stand in an aisle, browse on their phone, ask an AI assistant for a recommendation, or complete the purchase with a competitor. AI has become an accelerated new marketing channel demanding new strategies instantly. Price changes at the SKU level are making cost competition a precision science — and demand forecasting now requires hour-by-hour accuracy to catch waves and troughs. Even predictive supply chain management can be a deciding factor during disruptions.

When the prize of the opportunity is a potential 3–5% comp-sales growth from AI-enhanced data capabilities,[1] consumer firms have spent the last decade tapping into the power of AI across workflows from supply chain predictive management to customer recommendations to financial management.[2] And just as customers now expect assistance at the speed of AI models, retailers and CPG firms require the backend capabilities to support:

- Real-time inventory visibility and management
- Rock solid and predictive supply chain management
- Dialed-in demand forecasting
- Marketing systems that don't just respond changing behavior, but predict it

1 Databricks. January 2026, "Retail industry outcome map." (Internal document)
2 "Making AI Deliver," Databricks x The Economist, May 4, 2026

While aspirations of scaled AI across these functions remains high, a sharp bifurcation within retail and CPG firms suggests that the industrialization of the AI-native firm takes more than a general "applied AI." As industry leaders emerge, the point of differentiation is that they are not simply placing AI instruments in the hands of individual functions; they instead conduct them as a coordinated enterprise system. When merchandising, demand planning, supply chain, pricing, marketing, sales, and finance act from connected data and shared decisions, AI can create value greater than the sum of its use cases.

As expected, retailers have advanced fastest where data is already rich and connected, and decisions happen frequently, such as forecasting, replenishment, and pricing.

But it's no longer only data or AI silos that are a concern; now retailers are breaking down agentic silos for scaled efficiency and cross-enterprise AI learning. Every new architecture or third-party technology must have its own agent that's also connected to the wider "mind," sharing with each other and referencing the same secure, governed data.

According to recent research by BCG:

45% of retailers are scaling significant AI impact.

40% have barely begun.

In CPG, the arena is much starker: 75% of CPG companies remain in pilot/exploration.

Only 18% are scaling significant impact.[3]



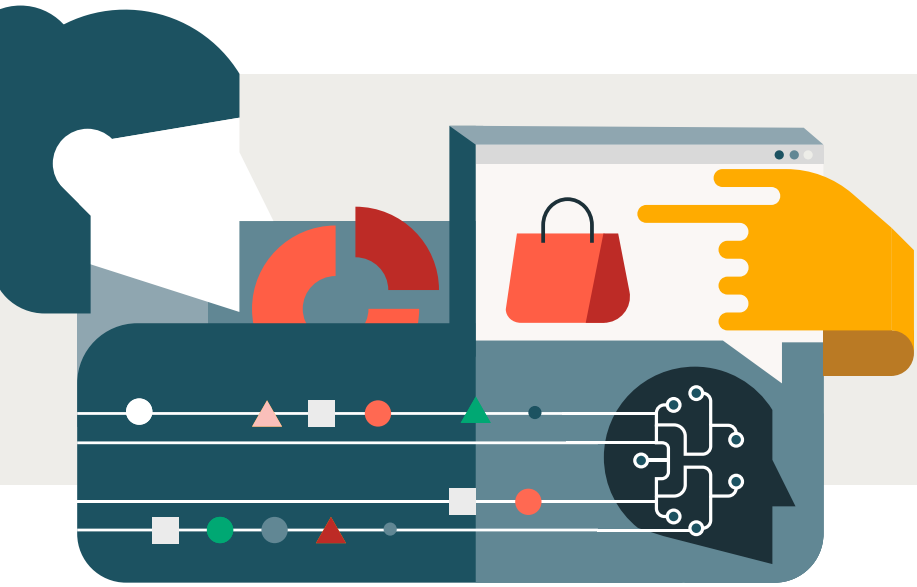
3 "AI in CPG and Retail: How Winners Are Pulling Ahead," By Mai-Britt Poulsen, Nicolas de Bellefonds, Yotam Ariav, Nate Shenck, Nick Goad, and Arnaud Bassoulet, BCH, June 24, 2026

Brands increasingly compete on their ability to sense, predict, and respond to those signals in real time at scale and on a one-to-one basis while responding to a simultaneous margin pressure as well. While studies have shown growing margins in 2026, new shopping behaviors, elevated inflation, and price-sensitive consumers are driving price competition.[4] As many as 81% of retail executives foresee margin expansion in the coming year.[5] However, supply chains are volatile; consumers expect more personalized and seamless experiences; and emerging agentic commerce raises the stakes further.

In response to these changing factors, fast-expanding agentic capabilities are transforming AI from recommender to consumer concierge. Shoppers are already relying on them to find the best products from the most trustworthy brands at the best prices.

It's no longer simply the presence of AI, or agent-readable sites that drives differentiation, but the context.

Databricks and Microsoft have been watching retail businesses closely and have refined the major success factors down to four critical criteria enterprises should use when evaluating new technologies and direction.



We call them the four C's:

- Context
- Control
- Cost
- Choice

4 "Retail Resilience Defies Doomsayers," John O'Trakoun, Federal Reserve Bank of Richmond, July 14, 2026

5 "2026 Retail Industry Global Outlook," Evan Sheehan, Brian McCarthy, Natalie Martini, Lupine Skelly, Dr. Bryn Walton, Oliver Vernon-Harcourt; Deloitte, January 8, 2026

1 Context: Can you trust AI to understand your business?

AI can know a lot about retail without knowing your retail business.

A frontier model may understand in general what “sell-through” or “loyalty tier” mean generically. But it needs your company’s governed, machine-readable definition of those terms — computed your way, on your data, ranked by your authority markers — to protect margin and drive growth. That institutional meaning is unique to your organization; no other development team can replicate and no competitor can copy it. AI must know what those terms mean not only within your unique company, but for each team and team member. A merchandiser has different context — and data needs — than a buyer.

In short: If it can’t inform you about the details of your business and its parts, how can you trust it at all? And if you can’t, how can your teams?

That’s how trust, and an AI advantage, can fade away.



If your AI and data don’t offer these answers specific to your company and its teams, it’s generic and useless:

- What constitutes a promotion?
- Which promotions are my unique customers most attracted to?
- How do I localize effective promotions to regions, currencies, and cultures?
- How can my merchandising strategy address demand shifting from online to in store orders?
- How should our customer tiers shift for new indicators of “loyal” customers?
- When product categories show return spikes how can R&D mitigate it?

If an AI agent doesn't know, or misinterprets any of these, it's not just a bad answer to a consumer; the agent may actually transact on the mistake or false understanding of the business or customer. That's a deeper problem. And the power of context extends far beyond definitions.

What looks like a slow sales week to a human appears different when AI points out that the same week last year included a product that went viral on TikTok. A demand forecast may be incomplete without weather. A pricing decision may need current freight costs, tariffs, rebates, inventory, promotional history, and margin thresholds in addition to last month's sales. But none of those signals help if they live in separate systems, arrive on different timelines, or lack a common definition the merchant and the agent both trust. Context becomes powerful when it's governed, connected, and fresh enough to act on.

The question worth asking is whether your data foundation was built for this. Many platforms were designed around curated summaries for dashboards and reports, but an agent deciding what to promote, price, or promise needs the full picture: SKU-level, store-level, hour-by-hour.

Azure Databricks was built around machine learning use cases before generative AI was the headline, which means it holds retail data at the grain agents actually need, governed and queryable, rather than summarized for a quarterly review.

That's the power of holistic business context. And critically, it isn't a one-time project. Not long ago, capturing these relationships meant manually building a knowledge graph, defining every entity and connection before the business got a single trustworthy answer. With Azure Databricks, the context layer builds and maintains itself, learning from the dashboards, queries, and reports your teams already produce, and getting richer as more of the business connects to it. When given that kind of governed, living context, AI can help make business decisions much faster than humans could on their own.

How Genie & Unity Catalog make context work

Define once, inherit everywhere

Traditional approaches to business context, whether data warehouses, BI semantic layers, or document retrieval, were built to inform humans looking at dashboards, not to ground an agent about to reprice a SKU or act on a promotion rule.

Azure Databricks solves this with two capabilities working together. Unity Catalog is where business definitions live once: what "on-promotion" means, how "comp sales" is calculated, which table is authoritative when two disagree. Every surface inherits those definitions, so the merchant's dashboard, the planner's forecast, the associate's mobile app, and the customer-facing agent all read from the same source of truth.

Genie puts that governed context into the hands of every role. A merchandiser can ask, "Which customer segments drove the most incremental revenue from last week's promotion?" and get a governed answer in seconds, because Genie inherits what those terms mean in your business from Unity Catalog. When definitions conflict, Genie's Ontology ranks by authority, favoring what's certified and clearly owned. And the context layer keeps improving on its own, learning from the queries and reports teams already produce. A planner can combine POS trends, weather, and freight costs to spot demand shifts earlier, because Genie draws from the same governed foundation.

> That's how you build trust, and drive a 20–30% faster response to demand changes.*



*Databricks. January 2026, "Retail industry outcome map." (Internal document)

The stakes rise even higher as retail becomes more agentic. In Adobe's global survey of 4,000 consumers, 56% expected AI to improve their customer experience, 49% thought it could save them money, and 46% expected more relevant recommendations. Nearly half (46%) said they do not care whether a brand uses AI, provided their needs are met.[6] As consumers evaluate every outcome of the AI applications of their favorite brands and retailers, a single sour customer-experience issue, whatever the cause — data, model, prompt error, or settings mistake — can have broader consequences.

CASE STUDY

7-Eleven — The locality of scale

The ubiquitous US convenience store found that the Genie foundation scales beautifully. 7-Eleven uses the Databricks Data + AI Platform to run a multipurpose agentic marketing assistant that tracks store-performance data across 13,000+ stores with compound AI systems and is standardizing on Unity Catalog — an agentic use case grounded in governed retail data at national scale. That's how intimate, specific context becomes applicable at scale.



6 "How the Rise of AI Is Changing Customer Behaviors," Adobe report, 2026

2 Control: The key to greater retail freedom

If context tells AI and agentic systems what's true, control determines what it's allowed to do about that truth.

Think about what happens today when a merchant spots a markdown opportunity. They pull the data, check the rules, run it past a manager, and make the call. The human is the trust check. If the pricing is wrong, the human must catch it before anything happens.

Now remove the human.

When an AI agent moves from answering questions to taking actions — approving a markdown, issuing a refund, pushing a promotion to a million loyalty members — the stakes change completely. A wrong answer from a rep requires an apology. An incorrect action nobody authorized becomes a liability — and can happen at scale in an agentic world.

This is the trust inversion that every retailer scaling AI will face: the more you automate, the more the system underneath has to carry the trust the human used to carry. Governance doesn't compete with automation ambition. It's what makes automation possible.

So what does real control look like in practice?

In an increasingly agentic retail environment, firms must have one governance plane, from data through models, agents, tools, and every connection an agent reaches.

With Azure Databricks, that plane is Unity Catalog — the same catalog that already governs business data now extends to govern AI activities. Additionally, Unity Gateway sits in front of every model and agent interaction, enforcing what an agent is allowed to do (not just what it can see), logging every action, and applying guardrails for PII, prompt injection, and domain-specific rules. When something goes wrong, there's a record of who — or what — authorized it.

That's what Adidas built. Its GenAI analyzes sentiment across millions of customer reviews, transforming unstructured feedback into product-innovation signals. But the reason it works at scale isn't the model — it's that the entire system operates on a governed foundation using Azure Databricks AI Search, Model Serving, Unity Catalog, and MLflow. The intelligence and the governance travel together.

The pattern holds across every retail use case. It's like a finance control tower answering permissioned questions across sales, margin, cost, and cash. Smart-store systems deliver next-best actions to associates on the floor; pricing agents operate within defined boundaries. These only become possible when the governance is already in place.

The retailers who will move fastest into agentic AI aren't the ones with the loosest rules. They're the ones who governed early enough that they can say yes — because they have the audit trail, the lineage, and the control plane that a regulator, a payment network, and their own board will accept.

More control doesn't mean less freedom. It means the freedom to let AI actually do the work.



CASE STUDY

Walgreens: Achieves near-real-time data deployment

Walgreens faced roughly a 48-hour delay before store data became usable, contributing to inventory miscalculations and stockouts. Using Azure Databricks, they unified supply chain and patient data to forecast demand and redistribute medications in near real time across nearly 9,000 locations, processing 40,000 data events per second.[7]

7 Databricks, "Azure Databricks for RCG." (Internal document)

3 Cost: Maximizing AI capabilities. Minimizing token costs.

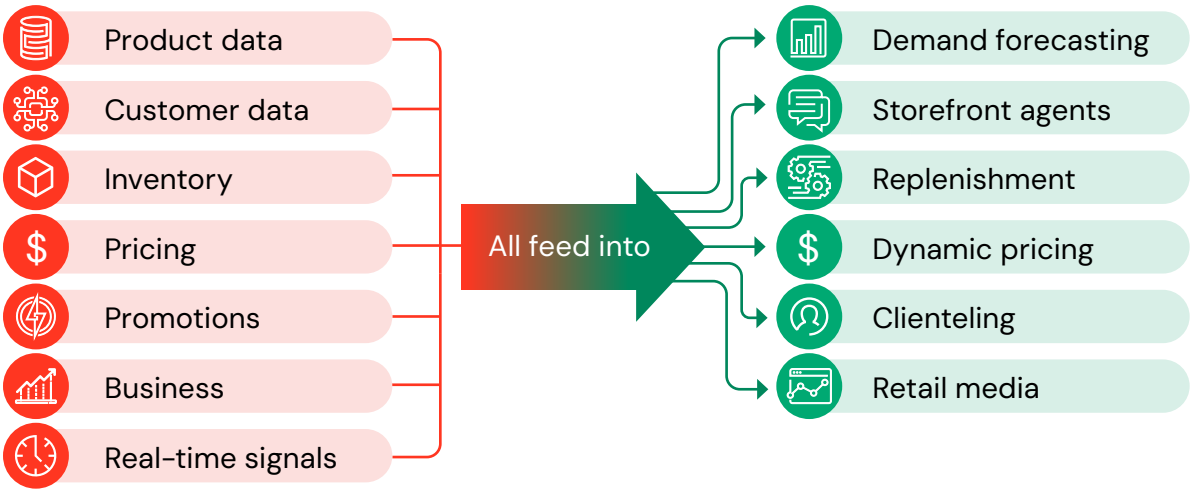
The economics of AI change when experimentation becomes infrastructure. A handful of pilots may require only a few models, datasets, pipelines, and queries. Production AI across every store, every SKU, and every planner and merchandiser in the company is a different animal — millions of automated decisions a week, each one consuming data, compute, and tokens. At that volume, every new use case becomes a cost multiplier unless it shares a foundation with the last one.

This is where reuse becomes a critical lever. Retailers already possess many of the ingredients required by multiple AI applications:

- Product and customer data
- Inventory positions
- Pricing and promotion histories
- Business definitions, governance policies, and real-time signals

A governed foundation like Azure Databricks allows you to prepare assets once and reuse them across multiple functions: demand forecasting, replenishment, clienteling, dynamic pricing, retail media, and customer-facing agents. All with the appropriate governance controls and context that matches the intended use. The alternative is paying to extract, embed, interpret, and govern the same inventory table six times for six applications.

Build once. Use many times.



Then there's the cost of the AI itself; more users, agents, models, and interactions inevitably mean more inference and token consumption, and that means more investment.

Within Azure Databricks, Unity Gateway routes every model and agent request through one control plane: spend attributed by team, application, and agent; budgets and rate limits set before a runaway holiday-season agent becomes a five-figure surprise; automatic routing to a smaller model when a smaller model will do. Finance sees what AI costs by use case instead of one line item labeled "AI."

Azure Databricks brings data, analytics, governance, and AI together on one platform, so the cost of the next use case is the marginal cost of the decision it produces — not the cost of standing up another stack. For a firm operating on low single-digit net margin, that gap decides whether an AI program protects margin or quietly eats it.

A unified AI control layer

At its core, Unity Gateway provides a secure, governed, and efficient way to manage the diverse and rapidly growing number of AI models and applications retail and CPG industries rely on. As these sectors increasingly use GenAI for everything from customer service to supply chain optimization, the Gateway offers a centralized layer to control costs, mitigate risks, and ensure consistent, high-quality outputs without being locked into a single AI provider.

4 Choice: Model agility means wider choice without migration

Think about how a retailer treats its supplier base. No experienced merchant would source an entire category from a single vendor; not because the vendor is bad, but because the market moves, leverage matters, and the best option this quarter may not be the best option next quarter. The same instinct applies to AI.

Model leadership changes constantly. Commerce protocols are forming around multiple competing approaches. The front door to shopping is being rebuilt around AI agents, and nobody yet knows which protocols will win. A platform decision made today shouldn't become a multiyear rebuilding project when the landscape shifts next quarter.

This is where open architecture earns its keep. Azure Databricks is built on open formats so your data stays yours regardless of which model, tool, or cloud you choose next. There's no proprietary format converting today's vendor preference into tomorrow's migration cost.

The same principle applies to how your teams interact with AI. Genie-powered experiences can run through Microsoft Teams, Copilot Studio agents, mobile, or custom web applications — all from the same governed foundation. You don't need a separate intelligence layer for every interface. The investment compounds across surfaces rather than being rebuilt for each one.

Making the store itself intelligent

The choice of technologies that feed AI applies to the aisles of the physical store as well. Connected-store systems can combine POS, cameras, IoT sensors, traffic, inventory, staffing, and equipment data. AI can flag an empty shelf, predict browsing patterns and checkout congestion, identify shrink patterns, anticipate refrigeration failure, or help determine staffing needs before those issues become lost sales or unhappy customers. The point isn't any single sensor or system. It's what brings them together. To do that the intelligence layer underneath them must be open, governed, and shared — all plugging into the same foundation rather than creating more silos.



The imperatives of open architecture

As AI agents and assistants increasingly become the front door to shopping, retailers will face a new question of strategy and technology: Who really owns the customer relationship?

Your first-party data — customer profiles unified across stores, e-commerce, mobile, loyalty, payment, and every other touchpoint — is the asset no frontier lab and no agent platform can replicate. A model maker can sell intelligence; it cannot sell you your own customers' purchase history, substitution preferences, loyalty behavior, and basket patterns. That privileged knowledge is the structural advantage, and it's yours only as long as you govern it.

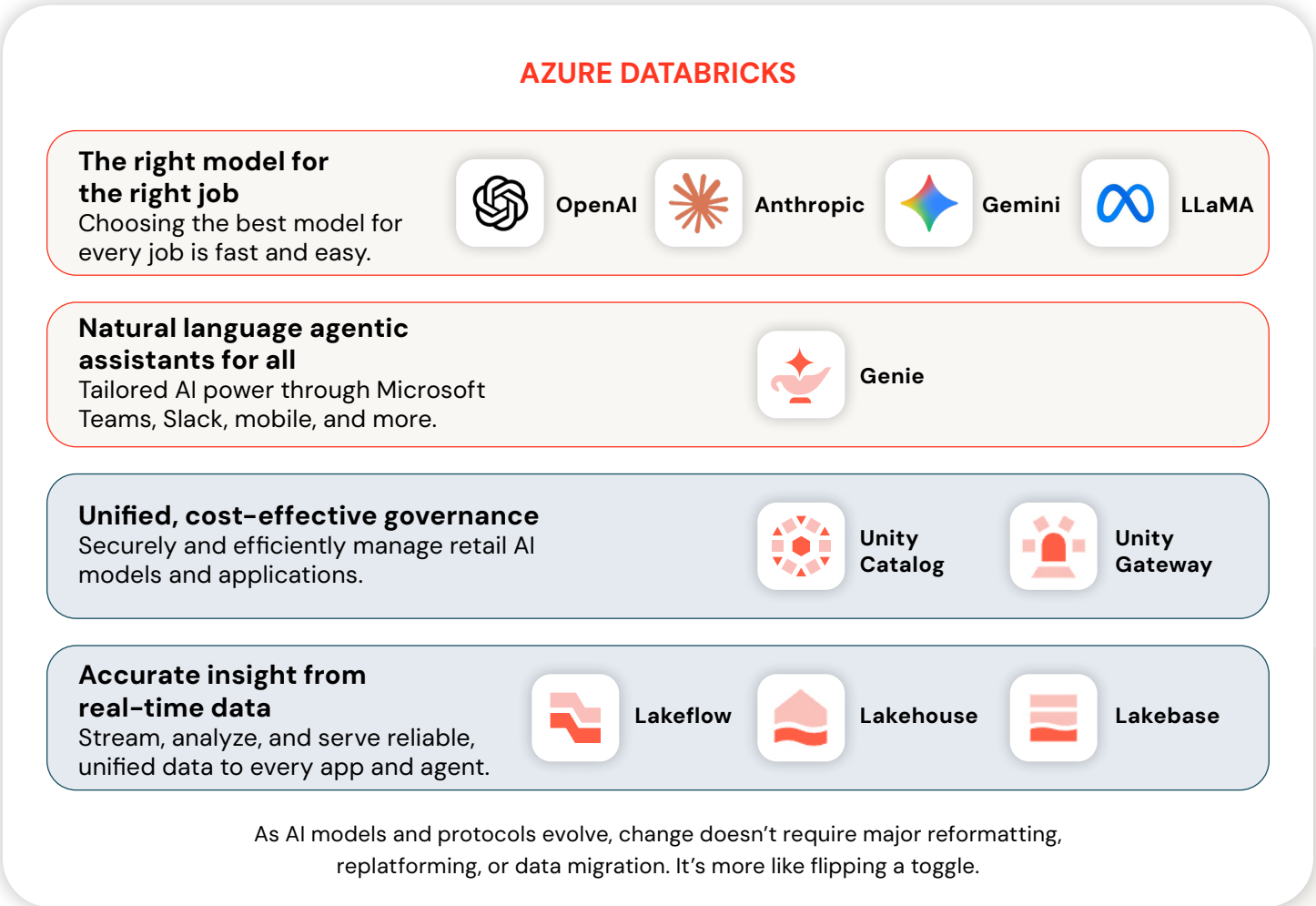
Open architecture is what makes that possible. When your data lives in open formats on your own infrastructure, governed by your own policies, you can participate in every new agent platform and commerce channel without surrendering the underlying asset. New agents become additional ways to reach the customer not a path to having the customer relationship owned by someone else.

That may be the most important kind of choice of all.

- **Keep the data.**
- **Keep the context.**
- **Keep the customer relationship.**
- **Choose the strategy.**



An architecture for agility and governance



As AI models and protocols evolve, change doesn't require major reformatting, replatforming, or data migration. It's more like flipping a toggle.

Across all these layers, the constant is your business vision and direction. That's the advantage of an open architecture. It helps prevent a model or platform preference from turning into a future conversion project or business strategy drift.

Azure Databricks supports multiple models while the retailer's governed first-party data remains independent of those choices.

More ways the four C's work for you

Individually, each of the four C's addresses a different requirement for scaling AI. Together, they create something more valuable: a foundation that supports AI for the everyday decisions that determine retail performance. And they're already driving results.

Predicting demand 20% to 30% faster[8]

A sudden change in weather, a product taking off on social media, supplier delays, localized demand spikes. AI demand forecasting can combine POS, inventory, search, social, weather, events, and other signals at the store/SKU level. This helps you detect demand changes 20–30% faster, while AI demand forecasting can improve forecast accuracy substantially.

8 Databricks. January 2026, "Retail industry outcome map." (Internal document)

Finding more margin where CFOs rarely look

There's a whole new realm where one executive decision can grow revenue and margin. While traditional blanket discounting treats every shopper, product, and store the same, even when they're not, AI doesn't have to. It can drive dynamic pricing that combines demand, competitive pricing, inventory, customer segments, and margin requirements. Markdown optimization can determine when — and how deeply — to discount aging inventory. Personalized promotions can target price-sensitive customers without unnecessarily discounting to shoppers likely to buy anyway. And the results can be astonishing:

15–25%
reductions in markdowns through optimal timing for profitable inventory clearance

30–40%
less promotional waste through personalized promotions

2–3%
margin improvement from private-label optimization

3–5%
comp-sales growth from assortment optimization[9]



USE CASE

Give every associate a better next move

An associate equipped with an AI-assisted mobile app can access current inventory, customer context, and product information while serving a shopper. AI task prioritization can simultaneously use real-time store conditions to direct associates toward the work that matters most. It's estimated that intelligent task management can reduce unproductive time by 30%.[10]

9 Databricks. January 2026, "Retail industry outcome map." (Internal document)
10 Ibid.

More than a partnership — a jointly-developed solution

Azure Databricks operates as a single, fully integrated vertical stack for retail and CPG jointly developed by Microsoft and Databricks. You control it as you control any Azure solution that's natively integrated with Microsoft services like Power BI, Azure OpenAI, and Microsoft Entra. So you get a single, highly controllable and auditable path from data to AI to business action built on the open Databricks Lakehouse foundation.

With Microsoft providing the trusted Azure operating environment, Databricks transforms that environment into a single data and AI engine for the enterprise. It's a three-layer structure that takes your operations from raw data to governed insight to production AI for tangible and measurable business impact.[1]



Three Layers. One retail powerhouse.



Model Access and AI Layer: Genie as the front door

- Genie gives users governed access to Databricks data from inside Microsoft Copilot, Teams, Copilot Cowork, and Microsoft 365.
- They can ask a question where they already work instead of leaving to go find it.
- Genie provides domain-aware analytics using enterprise data.



Built-in Governance Layer: Unity Catalog

- Unity Catalog governs every data asset, model, and AI workload.
- Security and access rules are set once and apply in Microsoft's tools and in Databricks so nothing falls out of sync.
- AI and sensitive data are combined so there's no need for parallel review processes for every use case.



Infrastructure layer

- Fast-moving retail data streams through Azure Event Hubs into Lakeflow.
- Schema and governance enforced by Unity Catalog from the moment it lands, not after.
- No need for pipeline rebuilds just to add a new tool — the format is universally readable.

No more shallow, contextless answers to complex enterprise data questions. The bottom-line business payoff: Less distance from question to insight, and from insight to action.

It’s all about the readiness

Neither AI progress nor growing retail customer expectations are slowing down. That makes readiness and agility more valuable than any single AI capability or market insight. With Context, Control, Cost, and Choice, consumer firms can keep evolving how you use AI to give you the flexibility to take advantage of what comes next. The opportunity isn’t simply to use more AI, it’s to make more of AI for consumer business ready for tomorrow’s marketplace.

Learn more about Azure Databricks for Retail

